

Linear Initial-Value Problem for the Finite-Depth Rayleigh–Taylor Instability

Abstract

This note summarizes the linear initial-value formulation implemented in the latest Mathematica notebook. Two inviscid, incompressible, irrotational fluids are separated by a nominally flat interface at $z = 0$ and are bounded by rigid horizontal walls at $z = H_1$ and $z = -H_2$. The initial interface is flat, but an interfacial vertical perturbation velocity $v_0(x)$ is imposed at $t = 0$. The resulting dynamics are solved mode by mode using the finite-depth Rayleigh–Taylor dispersion relation with surface tension.

1 Geometry and assumptions

The upper fluid occupies

$$0 < z < H_1, \quad (1)$$

while the lower fluid occupies

$$-H_2 < z < 0. \quad (2)$$

The unperturbed interface is located at $z = 0$. The densities are denoted by ρ_1 for the upper fluid and ρ_2 for the lower fluid. The gravitational acceleration is g , and the surface tension coefficient is σ .

The interface displacement is denoted by

$$z = \eta(x, t), \quad (3)$$

where η is assumed sufficiently small for linearization. The rigid-wall impermeability conditions are

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \quad (4)$$

where w_1 and w_2 are the vertical velocities in the upper and lower fluids, respectively.

The initial-value problem considered in the notebook is

$$\eta(x, 0) = 0, \quad \partial_t \eta(x, 0) = v_0(x). \quad (5)$$

Thus, the interface is initially flat, and the perturbation is introduced through the initial vertical velocity at the interface.

2 Finite-depth dispersion relation

For a Fourier mode with horizontal wavenumber $k > 0$, the corrected finite-depth dispersion relation used in the notebook is

$$\omega^2(k) = \frac{k(\rho_1 - \rho_2)g - \sigma k^3}{\rho_1 \coth(kH_1) + \rho_2 \coth(kH_2)}. \quad (6)$$

The sign of the capillary term is stabilizing: for sufficiently large k , the term $-\sigma k^3$ can make $\omega^2(k)$ negative, producing stable oscillatory modes rather than exponential growth.

When $\rho_1 > \rho_2$, the critical wavenumber at which the numerator changes sign is

$$k_c = \sqrt{\frac{(\rho_1 - \rho_2)g}{\sigma}}. \quad (7)$$

For $0 < k < k_c$, the numerator of (6) is positive, and the corresponding modes are Rayleigh–Taylor unstable. For $k > k_c$, surface tension stabilizes the mode in the linear theory.

3 Modal initial-value problem

Consider a periodic domain of length L . The horizontal wavenumbers are

$$k_n = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}. \quad (8)$$

The zero mode is omitted in the notebook, since the instability calculation concerns nonzero interfacial perturbations. The initial velocity is expanded as

$$v_0(x) = \sum_{n \neq 0} \hat{v}_n e^{ik_n x}, \quad (9)$$

with Fourier coefficients

$$\hat{v}_n = \frac{1}{L} \int_0^L v_0(x) e^{-ik_n x} dx. \quad (10)$$

For each nonzero mode, define

$$\lambda_n = \omega^2(|k_n|). \quad (11)$$

The modal interface amplitude satisfies

$$\frac{d^2 \hat{\eta}_n}{dt^2} = \lambda_n \hat{\eta}_n, \quad (12)$$

with initial conditions

$$\hat{\eta}_n(0) = 0, \quad \frac{d\hat{\eta}_n}{dt}(0) = \hat{v}_n. \quad (13)$$

The solution is

$$\hat{\eta}_n(t) = \hat{v}_n G(\lambda_n, t), \quad (14)$$

where

$$G(\lambda, t) = \begin{cases} \frac{\sinh(\sqrt{\lambda} t)}{\sqrt{\lambda}}, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \quad (15)$$

Its time derivative is

$$\partial_t G(\lambda, t) = \begin{cases} \cosh(\sqrt{\lambda} t), & \lambda \neq 0, \\ 1, & \lambda = 0. \end{cases} \quad (16)$$

If $\lambda < 0$, these formulae are interpreted through complex arguments. Equivalently, writing $\lambda = -\alpha^2$ with $\alpha > 0$, one obtains

$$G(-\alpha^2, t) = \frac{\sin(\alpha t)}{\alpha}, \quad \partial_t G(-\alpha^2, t) = \cos(\alpha t), \quad (17)$$

which corresponds to stable capillary-gravity oscillations.

The reconstructed interface displacement is

$$\eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{\eta}_n(t) e^{ik_n x} \right]. \quad (18)$$

The interfacial vertical velocity is

$$w(x, 0, t) = \partial_t \eta(x, t) = \text{Re} \left[\sum_{n \neq 0} \hat{v}_n \partial_t G(\lambda_n, t) e^{ik_n x} \right]. \quad (19)$$

4 Vertical velocity fields

The finite-depth vertical structure used in the notebook satisfies the impermeability conditions at the two walls. If the interfacial vertical velocity amplitude of mode n is denoted by

$$\hat{w}_n(0, t) = \hat{v}_n \partial_t G(\lambda_n, t), \quad (20)$$

then the upper-fluid vertical velocity is reconstructed using

$$w_{1,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)}. \quad (21)$$

Similarly, the lower-fluid vertical velocity is

$$w_{2,n}(z, t) = \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)}. \quad (22)$$

Therefore,

$$w_1(x, z, t) = \text{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(H_1 - z))}{\sinh(|k_n|H_1)} e^{ik_n x} \right], \quad (23)$$

for $0 < z < H_1$, and

$$w_2(x, z, t) = \text{Re} \left[\sum_{n \neq 0} \hat{w}_n(0, t) \frac{\sinh(|k_n|(z + H_2))}{\sinh(|k_n|H_2)} e^{ik_n x} \right], \quad (24)$$

for $-H_2 < z < 0$.

These formulae immediately imply

$$w_1(x, H_1, t) = 0, \quad w_2(x, -H_2, t) = 0, \quad (25)$$

and

$$w_1(x, 0, t) = w_2(x, 0, t) = \partial_t \eta(x, t). \quad (26)$$

5 Example used in the notebook

The notebook uses the parameter set

$$\rho_1 = 2, \quad \rho_2 = 1, \quad g = 9.81, \quad \sigma = 0.01, \quad H_1 = H_2 = 1, \quad L = 2\pi. \quad (27)$$

The imposed initial interfacial velocity is

$$v_0(x) = U_0 \cos(3x) + \frac{2}{5} U_0 \sin(5x), \quad U_0 = 10^{-3}. \quad (28)$$

Since $L = 2\pi$, the nonzero exact Fourier coefficients are

$$\hat{v}_3 = \frac{U_0}{2}, \quad \hat{v}_{-3} = \frac{U_0}{2}, \quad (29)$$

$$\hat{v}_5 = -\frac{iU_0}{5}, \quad \hat{v}_{-5} = \frac{iU_0}{5}. \quad (30)$$

The numerical notebook nevertheless computes the coefficients through a discrete periodic projection, which is more general and can be used for arbitrary smooth periodic profiles.

6 Discrete Fourier projection in the Mathematica notebook

The notebook uses a uniform grid

$$x_m = \frac{mL}{N}, \quad m = 0, 1, \dots, N-1, \quad (31)$$

where $N = 4096$. The Fourier coefficient is approximated by the periodic trapezoidal rule

$$\hat{v}_n \approx \frac{1}{N} \sum_{m=0}^{N-1} v_0(x_m) e^{-ik_n x_m}. \quad (32)$$

This avoids the convergence warnings caused by numerical quadrature when the exact coefficient is zero. The Mathematica implementation also applies **Chop** to remove round-off residuals.

7 Verification conditions

The notebook verifies the initial conditions by computing

$$\max_x |\eta(x, 0)|, \quad \max_x |\partial_t \eta(x, 0) - v_0(x)|. \quad (33)$$

For a sufficiently resolved Fourier truncation, both quantities should be close to machine precision.

The wall boundary conditions are verified by computing

$$\max_x |w_1(x, H_1, 0)|, \quad \max_x |w_2(x, -H_2, 0)|. \quad (34)$$

Because the vertical shape functions vanish exactly at the corresponding rigid walls, these errors should also be at the level of numerical round-off.

A Core Mathematica implementation

The following listing contains the essential part of the corrected Mathematica notebook.

```
Remove["Global`*"];

pars = {
  rho1 -> 2.0,
  rho2 -> 1.0,
  g -> 9.81,
  sigma -> 0.01,
  H1 -> 1.0,
  H2 -> 1.0,
  L -> 2 Pi
};
```

```

Lnum = N[L /. pars];

omega2[k_] := (k (rho1 - rho2) g - sigma k^3)/
  (rho1 Coth[k H1] + rho2 Coth[k H2]);

omega2Num[k_?NumericQ] := N[omega2[k] /. pars];

kCritical = Sqrt[((rho1 - rho2) g)/sigma] /. pars;

tol = 10^-12;

growthKernel[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    t,
    Sinh[Sqrt[lambda] t]/Sqrt[lambda]
  ];

growthKernelDot[lambda_?NumericQ, t_?NumericQ] :=
  If[Abs[lambda] < tol,
    1,
    Cosh[Sqrt[lambda] t]
  ];

U0 = 10^-3;

v0Expr = U0 Cos[3 x] + (2/5) U0 Sin[5 x];

v0[xVal_?NumericQ] := N[v0Expr /. x -> xVal];

nMax = 32;

kn[n_Integer] := 2 Pi n/Lnum;

modes = Join[Range[-nMax, -1], Range[1, nMax]];

nSamples = 4096;

xGrid = N[Lnum Range[0, nSamples - 1]/nSamples];
v0Grid = v0 /@ xGrid;

Clear[vHat];

vHat[n_Integer] := vHat[n] =
  Chop[
    (1/nSamples) Total[v0Grid Exp[-I kn[n] xGrid]],
    10^-14
  ];

etaHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernel[omega2Num[Abs[kn[n]]], tVal];

eta[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      etaHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

```

```

wInterfaceHat[n_Integer, tVal_?NumericQ] :=
  vHat[n] growthKernelDot[omega2Num[Abs[kn[n]]], tVal];

wInterface[xVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      wInterfaceHat[#, tVal] Exp[I kn[#] xVal] & /@ modes
    ]
  ];

shape1[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k ((H1 /. pars) - zVal)]/Sinh[k (H1 /. pars)];

shape2[k_?NumericQ, zVal_?NumericQ] :=
  Sinh[k (zVal + (H2 /. pars))]/Sinh[k (H2 /. pars)];

w1[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      (wInterfaceHat[#, tVal]
       shape1[Abs[kn[#]], zVal]
       Exp[I kn[#] xVal]) & /@ modes
    ]
  ];

w2[xVal_?NumericQ, zVal_?NumericQ, tVal_?NumericQ] :=
  Re[
    Total[
      (wInterfaceHat[#, tVal]
       shape2[Abs[kn[#]], zVal]
       Exp[I kn[#] xVal]) & /@ modes
    ]
  ];

checkGrid = N[Lnum Range[0, 512]/512];

maxEtaInitialError =
  Max[Abs[eta[#, 0] & /@ checkGrid]];

maxVelocityInitialError =
  Max[Abs[(wInterface[#, 0] - v0[#]) & /@ checkGrid]];

maxUpperWallError =
  Max[Abs[w1[#, H1 /. pars, 0] & /@ checkGrid]];

maxLowerWallError =
  Max[Abs[w2[#, -(H2 /. pars), 0] & /@ checkGrid]];

{
  maxEtaInitialError,
  maxVelocityInitialError,
  maxUpperWallError,
  maxLowerWallError
}

```